

Bayesian Linear Models

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Linear Regression

- Linear regression is, perhaps, *the* most widely used statistical modeling tool.
- It addresses the following question: How does a quantity of primary interest, y , vary as (depend upon) another quantity, or set of quantities, x ?
- The quantity y is called the *response* or *outcome variable*. Some people simply refer to it as the *dependent variable*.
- The variable(s) x are called *explanatory variables*, *covariates* or simply *independent variables*.
- In general, we are interested in the conditional distribution of y , given x , parametrized as $p(y \mid \theta, x)$.

- Typically, we have a set of *units* or *experimental subjects* $i = 1, 2, \dots, n$.
- For each of these units we have measured an outcome y_i and a set of explanatory variables $x_i^\top = (1, x_{i1}, x_{i2}, \dots, x_{ip})$.
- The first element of $x_i^\top$ is often taken as 1 to signify the presence of an “intercept”.
- We collect the outcome and explanatory variables into an $n \times 1$ vector and an $n \times (p + 1)$ matrix:

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} ; \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix} = \begin{pmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{pmatrix} .$$

- The linear model is the most fundamental of all serious statistical models underpinning:
 - ANOVA: y_i is continuous, x_{ij} 's are *all* categorical
 - REGRESSION: y_i is continuous, x_{ij} 's are continuous
 - ANCOVA: y_i is continuous, x_{ij} 's are continuous for some j and categorical for others.

Conjugate Bayesian Linear Regression: The M&m formula

- A conjugate Bayesian linear model is given by:

$$y_i | \mu_i, \sigma^2, X \stackrel{\text{ind}}{\sim} N(\mu_i, \sigma^2); \quad i = 1, 2, \dots, n;$$

$$\mu_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = x_i^\top \beta; \quad \beta = (\beta_0, \beta_1, \dots, \beta_p)^\top;$$

$$\beta | \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0); \quad \sigma^2 \sim IG(a_0, b_0).$$

- Unknown parameters include the regression parameters and the variance, i.e. $\theta = \{\beta, \sigma^2\}$.
- We assume X is observed without error and all inference is conditional on X .
- The above model is often written in terms of the posterior density $p(\theta | y) \propto p(\theta, y)$:

$$IG(\sigma^2 | a_0, b_0) \times N(\beta | M_0 m_0, \sigma^2 M_0) \times \prod_{i=1}^n N(y_i | x_i^\top \beta, \sigma^2).$$

Conjugate Bayesian (General) Linear Regression

- A more general conjugate Bayesian linear model:

$$y \mid \beta, \sigma^2, X \sim N(X\beta, \sigma^2 V_y)$$

$$\beta \mid \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0) ;$$

$$\sigma^2 \sim IG(a_0, b_0) .$$

- V_y , m_0 , M_0 , a_0 and b_0 are assumed fixed.
- Unknown parameters include the regression parameters and the variance, i.e. $\theta = \{\beta, \sigma^2\}$.
- We assume X is observed without error and all inference is conditional on X .
- The posterior density $p(\theta \mid y) \propto p(\theta, y)$:

$$IG(\sigma^2 \mid a_0, b_0) \times N(\beta \mid M_0 m_0, \sigma^2 M_0) \times N(y \mid X\beta, \sigma^2 V_y)$$

- The model on the previous slide is a special case with $V_y = I_n$ ($n \times n$ identity matrix).

Deriving $p(\sigma^2 | y)$

$$\begin{aligned} p(\sigma^2 | y) &= \int p(\sigma^2, \beta | y) d\beta \\ &= IG(\sigma^2 | a_0, b_0) \times \int N(\beta | M_0 m_0, \sigma^2 M_0) \times N(y | X\beta, \sigma^2 V_y) d\beta \\ &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2} \right)^{\frac{n+p}{2}} \int \exp \left\{ -\frac{1}{2\sigma^2} (Q_1(\beta) + Q_2(\beta, y)) \right\} , \\ &\hspace{25em} (1) \end{aligned}$$

where

$$\begin{aligned} Q_1(\beta) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) \quad \text{and} \\ Q_2(\beta, y) &= (y - X\beta)^\top V_y^{-1} (y - X\beta) . \end{aligned}$$

Deriving $p(\sigma^2 | y)$

$$\begin{aligned} Q_1(\beta) + Q_2(\beta, y) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) + (y - X\beta)^\top V_y^{-1} (y - X\beta) \\ &= \beta^\top M_0^{-1} \beta - 2\beta^\top m_0 + m_0^\top M_0 m_0 \\ &\quad + \beta^\top X^\top V_y^{-1} X \beta - 2\beta^\top X^\top V_y^{-1} y + y^\top V_y^{-1} y \\ &= \beta^\top (M_0^{-1} + X^\top V_y^{-1} X) \beta - 2\beta^\top (m_0 + X^\top V_y^{-1} y) \\ &\quad + m_0^\top M_0 m_0 + y^\top V_y^{-1} y \\ &= \beta^\top M^{-1} \beta - 2\beta^\top m + c \end{aligned} \tag{2}$$

where M is a symmetric positive definite matrix, m is a vector and c is a scalar given by:

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top V_y^{-1} X ; \\ m &= m_0 + X^\top V_y^{-1} y ; \\ c &= m_0^\top M_0 m_0 + y^\top V_y^{-1} y . \end{aligned}$$

Note: M , m and c do not depend upon β .

Deriving $p(\sigma^2 | y)$ (contd.)

Multivariate Completion of Square (ellipsoidal rectification)

$$\beta^\top M^{-1} \beta - 2\beta^\top m = (\beta - Mm)^\top M^{-1} (\beta - Mm) - m^\top Mm \quad (3)$$

Easy to verify by simply expanding the right hand side.

Another way (assuming you do not know the form on the right hand side) is to assume a centered quadratic form for the right hand side and find the coefficients. That is, write

$$\beta^\top M^{-1} \beta - 2\beta^\top m = (\beta - a)^\top B (\beta - a) + d ,$$

and try to find a (vector), B (matrix, assumed symmetric) and d (scalar) by comparing the coefficients with the left hand side. So, if the above is true then

$$\beta^\top M^{-1} \beta - 2\beta^\top m = \beta^\top B \beta - 2\beta^\top Ba + a^\top Ba + d .$$

Hence, we conclude:

$$B = M^{-1} ; Ba = m \implies a = B^{-1}m = Mm ; a^\top Ba + d = 0 \implies d = -a^\top Ba = -m^\top Mm .$$

Deriving $p(\sigma^2 | y)$ (contd.)

- Apply (3) to (2):

$$\begin{aligned} Q_1(\beta) + Q_2(\beta, y) &= (\beta - Mm)^\top M^{-1}(\beta - Mm) - m^\top Mm + c \\ &= (\beta - Mm)^\top M^{-1}(\beta - Mm) + c^*, \end{aligned} \quad (4)$$

where

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top V_y^{-1} X ; \\ m &= m_0 + X^\top V_y^{-1} y ; \\ c &= m_0^\top M_0 m_0 + y^\top V_y^{-1} y ; \\ c^* &= c - m^\top Mm = m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top Mm . \end{aligned}$$

Deriving $p(\sigma^2 | y)$ (concludes)

- Apply (4) to (1):

$$\begin{aligned} p(\sigma^2 | y) &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2}\right)^{\frac{n+p}{2}} \int \exp \left\{ -\frac{1}{2\sigma^2} (Q_1(\beta) + Q_2(\beta, y)) \right\} \\ &\propto IG(\sigma^2 | a_0, b_0) \times \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \int \left(\frac{1}{\sigma^2}\right)^{\frac{p}{2}} \exp \left\{ -\frac{1}{2\sigma^2} (\beta - Mm)^\top M^{-1} (\beta - Mm) + c^* \right\} \\ &\propto \left(\frac{1}{\sigma^2}\right)^{a_0+1} \exp \left\{ -\frac{b_0}{\sigma^2} \right\} \times \left(\frac{1}{\sigma^2}\right)^{\frac{n}{2}} \exp \left\{ -\frac{c^*}{2\sigma^2} \right\} \times \int N(\beta | Mm, \sigma^2 M) \\ &\propto \left(\frac{1}{\sigma^2}\right)^{a_0+\frac{n}{2}+1} \exp \left\{ -\frac{1}{\sigma^2} \left(b_0 + \frac{c^*}{2} \right) \right\} . \end{aligned}$$

- Therefore,

$$p(\sigma^2 | y) = IG(\sigma^2 | a^*, b^*)$$

where

$$a^* = a_0 + \frac{n}{2} ;$$

$$b^* = b_0 + \frac{c^*}{2} = b_0 + \frac{1}{2} \{ m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top Mm \}$$

Conjugate Bayesian (General) Linear Regression

- The joint posterior density can be written as

$$p(\beta, \sigma^2 | y) \propto \underbrace{IG(\sigma^2 | a^*, b^*)}_{p(\sigma^2 | y)} \times \underbrace{N(\beta | Mm, \sigma^2 M)}_{p(\beta | \sigma^2, y)},$$

where

$$a^* = a_0 + \frac{n}{2}; \quad b^* = b_0 + \frac{1}{2} (m_0^\top M_0 m_0 + y^\top V_y^{-1} y - m^\top M m);$$

$$m = m_0 + X^\top V_y^{-1} y; \quad M^{-1} = M_0^{-1} + X^\top V_y^{-1} X.$$

Algorithm Exact posterior sampling from $p(\beta, \sigma^2 | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: **end for**
 - 5: **return** $(\sigma_{(i)}^2, \beta_{(i)})_{i=1,2,\dots,N}$
-

Bayesian predictions from linear regression

- Let $\tilde{y}$ denote an $m \times 1$ vector of outcomes we seek to predict based upon predictors $\tilde{X}$.
- We wish to draw samples from:

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta, y) p(\theta | y) d\theta$$

- Or, equivalently, draw from joint $p(\theta, \tilde{y} | y)$, using composition and save samples of $\tilde{y}$:

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

```
1: for ( $i$  in  $1 : N$ ), do  
2:   Draw  $\theta_{(i)} \sim p(\theta | y)$   
3:   Draw  $\tilde{y}_{(i)} \sim p(\tilde{y} | \theta_{(i)}, y)$   
4: end for  
5: return  $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$ 
```

Bayesian predictions from linear regression (contd.)

- For legitimate probabilistic predictions (forecasting), the conditional distribution $p(\tilde{y} | \theta, y)$ must be well-defined.
- For example, consider the case with $V_y = I_n$. Specify the linear model:

$$\begin{bmatrix} y \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} \beta + \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} ; \quad \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} I_n & O \\ O & I_m \end{bmatrix} \right) .$$

- Easy to derive the conditional density:

$$p(\tilde{y} | \theta, y) = p(\tilde{y} | \theta) = N(\tilde{y} | \tilde{X}\beta, \sigma^2 I_m)$$

- Posterior predictive density:

$$p(\tilde{y} | y) = \int N(\tilde{y} | \tilde{X}\beta, \sigma^2 I_m) p(\beta, \sigma^2 | y) d\beta d\sigma^2 .$$

Bayesian predictions from linear regression (algorithm)

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: Draw $\tilde{y}_{(i)} \sim N(\tilde{X}\beta_{(i)}, \sigma_{(i)}^2 I_m)$
 - 5: **end for**
 - 6: **return** $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$
-

Bayesian predictions from general linear regression

- For example, consider the case with general V_y . Specify:

$$\begin{bmatrix} y \\ \tilde{y} \end{bmatrix} = \begin{bmatrix} X \\ \tilde{X} \end{bmatrix} \beta + \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} ; \quad \begin{bmatrix} \epsilon \\ \tilde{\epsilon} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} V_y & V_{y\tilde{y}} \\ V_{y\tilde{y}}^\top & V_{\tilde{y}} \end{bmatrix} \right) .$$

- Derive the conditional density $p(\tilde{y} | \theta, y) = N(\tilde{y} | \mu_{\tilde{y}|y}, \sigma^2 V_{\tilde{y}|y})$:

$$\mu_{\tilde{y}|y} = \tilde{X}\beta + V_{y\tilde{y}}^\top V_y^{-1}(y - X\beta) ; \quad V_{\tilde{y}|y} = V_{\tilde{y}} - V_{y\tilde{y}}^\top V_y^{-1} V_{y\tilde{y}} .$$

- Posterior predictive density:

$$p(\tilde{y} | y) = \int N(\tilde{y} | \mu_{\tilde{y}|y}, \sigma^2 V_{\tilde{y}|y}) p(\beta, \sigma^2 | y) d\beta d\sigma^2 .$$

Bayesian predictions from general linear regression (algorithm)

Algorithm Exact posterior predictive sampling from $p(\tilde{y} | y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\sigma_{(i)}^2 \sim IG(a^*, b^*)$
 - 3: Draw $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$
 - 4: Compute $\mu_{\tilde{y}|y}$ using $\beta_{(i)}$ and draw $\tilde{y}_{(i)} \sim N(\mu_{\tilde{y}|y}, \sigma_{(i)}^2 V_{\tilde{y}})$
 - 5: **end for**
 - 6: **return** $(\tilde{y}_{(i)})_{i=1,2,\dots,N}$
-

Bayesian Regression Analysis for BIG DATA

Bayesian updating with independent data

- Sequential update for $\theta = \{\beta, \sigma^2\}$: Update posterior for $t = 1, 2, \dots, T$

$$\begin{aligned} p(\theta \mid y_1, \dots, y_t) &\propto p(\theta, y_1, \dots, y_{t-1}, y_t) \\ &\propto \underbrace{p(\theta, y_1, \dots, y_{t-1})}_{p(y_1, \dots, y_{t-1}) \times p(\theta \mid y_1, \dots, y_{t-1})} \times \underbrace{p(y_t \mid \theta, y_1, \dots, y_{t-1})}_{p(y_t \mid \theta)} \\ &\propto p(\theta \mid y_1, \dots, y_{t-1}) \times p(y_t \mid \theta) . \end{aligned}$$

- Note the use of independent data given θ :

$$p(y_t \mid \theta, y_1, \dots, y_{t-1}) = p(y_t \mid \theta)$$

- Posterior at time $t - 1$ becomes the prior for the next t .
- Divide & Conquer by loading manageable blocks of data for each t .

Bayesian regression using Divide and Conquer

- Partition data as $\{y_t, X_t\}$, $t = 1, 2, \dots, T$, y_t is $n_t \times 1$, X_t is $n_t \times p$ and $n = \sum_{t=1}^T n_t$.
- Suppose we assume block independence:

$$y_t | \beta, \sigma^2 \stackrel{\text{ind}}{\sim} N(X_t \beta, \sigma^2 V_t);$$
$$\beta | \sigma^2 \sim N(M_0 m_0, \sigma^2 M_0); \quad \sigma^2 \sim IG(a_0, b_0) .$$

- Let $\theta = \{\beta, \sigma^2\}$. The posterior is $p(\theta | y) \propto p(\theta, y)$:

$$IG(\sigma^2 | a_0, b_0) \times N(\beta | M_0 m_0, \sigma^2 M_0) \times \prod_{t=1}^T N(y_t | X_t \beta, \sigma^2 V_t) .$$

- Note: If each data point is independent, then $V_t = I_{n_t}$. Otherwise, we can try to design data collection in a manner that data are independent across blocks.

Bayesian regression using Divide and Conquer

- Write the posterior distribution at time $t - 1$ as

$$p(\theta | y_1, \dots, y_{t-1}) = IG(\sigma^2 | a_{t-1}, b_{t-1}) \times N(\beta | M_{t-1}m_{t-1}, \sigma^2 M_{t-1})$$

- Posterior at time t updates as

$$p(\theta | y_1, \dots, y_t) = IG(\sigma^2 | a_t, b_t) \times N(\beta | M_t m_t, \sigma^2 M_t)$$

where a_t , b_t , m_t and M_t are updated as follows:

$$m_t = m_{t-1} + X_t^\top V_t^{-1} y_t; \quad M_t^{-1} = M_{t-1}^{-1} + X_t^\top V_t^{-1} X_t$$

$$a_t = a_{t-1} + \frac{n_t}{2}; \quad b_t = b_{t-1} + \frac{q_t}{2}, \quad \text{where}$$

$$q_t = y_t^\top y_t + m_{t-1}^\top M_{t-1} m_{t-1} - m_t^\top M_t m_t.$$

Bayesian analysis for BIG DATA (algorithm)

Algorithm Exact posterior sampling for Divide & Conquer

```
1: Set  $m \leftarrow m_0$ ,  $M^{-1} \leftarrow M_0^{-1}$ ,  $a \leftarrow a_0$ ,  $b \leftarrow b_0$ 
2:  $M \leftarrow \text{solve}(M^{-1}, I_p)$ 
3: for ( $t$  in  $1 : T$ ), do
4:    $q \leftarrow \|y_t\|^2 + m^\top M m$ 
5:    $m \leftarrow m + X_t^\top V_t^{-1} y_t$  and  $M^{-1} \leftarrow M^{-1} + X_t^\top V_t^{-1} X_t$ 
6:    $q \leftarrow q - m^\top M m$ 
7:    $b \leftarrow b + q/2$  and  $a \leftarrow a + n_t/2$ 
8: end for
9:
10: for ( $i$  in  $1 : N$ ), do
11:   Draw  $\sigma_{(i)}^2 \sim IG(a, b)$ 
12:   Draw  $\beta_{(i)} \sim N(Mm, \sigma_{(i)}^2 M)$ 
13: end for
14: return  $\left\{ \beta_{(i)}, \sigma_{(i)}^2 \right\}_{i=1, \dots, N}$ 
```

Bayesian analysis for BIG DATA (algorithm using linear algebra)

Algorithm Exact posterior sampling for Divide & Conquer

```
1: Set  $m \leftarrow m_0$ ,  $M^{-1} \leftarrow M_0^{-1}$ ,  $a \leftarrow a_0$ ,  $b \leftarrow b_0$ 
2:  $L^{-1} \leftarrow \text{cholesky}(M^{-1})$  and  $L = \text{trsolve}(L^{-1}, I_p)$ 
3: for ( $t$  in  $1 : T$ ), do
4:    $q \leftarrow \|y_t\|^2 + \|Lm\|^2$ 
5:    $m \leftarrow m + X_t^\top V_t^{-1} y_t$  and  $M^{-1} \leftarrow M^{-1} + X_t^\top V_t^{-1} X_t$ 
6:    $L^{-1} \leftarrow \text{cholesky}(M^{-1})$  and  $L = \text{trsolve}(L^{-1}, I_p)$ 
7:    $q \leftarrow q - \|Lm\|^2$ 
8:    $b \leftarrow b + q/2$  and  $a \leftarrow a + n_t/2$ 
9: end for
10:
11: for ( $i$  in  $1 : N$ ), do
12:   Draw  $\sigma_{(i)}^2 \sim IG(a, b)$  and  $z \sim N(0, I_p)$ 
13:    $\beta_{(i)} \leftarrow L^\top Lm + \sigma_{(i)} L^\top z$ 
14: end for
15: return  $\left\{ \beta_{(i)}, \sigma_{(i)}^2 \right\}_{i=1, \dots, N}$ 
```